

Implicit multiparameter full waveform inversion of multioffset ground penetrating radar data

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SUMMARY

Full waveform inversion (FWI) of ground-penetrating radar (GPR) data is one of the key methods for reconstructing high-resolution near-surface images and understanding changes in the near-surface environment. However, FWI faces great challenges due to the ill-posed nature of the optimization problem and its high sensitivity to the quality of the initial model. In this paper, we propose the implicit full waveform inversion (IFWI) algorithm for GPR that addresses these limitations by using an implicit and continuous function built by deep neural networks to represent multiple subsurface parameters. In this way, IFWI can be thought of as training a physically constrained neural network representation, allowing it to exploit the frequency principle in deep learning optimization, that is first recovering the low-frequency components of the subsurface parameters and large-scale structures, and then their high-frequency and fine-grained information, which makes IFWI an automated multiscale inversion approach without manual frequency band selection. Numerical experiments are conducted to verify the effectiveness and performance of IFWI relative to FWI. The results demonstrate that IFWI can accurately reconstruct subsurface images and anomalies even with a suboptimal initial model, while FWI and multiscale FWI struggle to provide useful information due to being trapped in local minima. Consequently, IFWI exhibits less dependence on the initial model compared to FWI and multiscale FWI. To better understand the behaviour of IFWI and its sensitivity to neural network architecture, we comprehensively discuss the influences of these hyperparameters on IFWI, providing initial selection guidelines and practical guidance for adjustments.

Key words: Ground penetrating radar (GPR); Machine learning; Neural networks; Waveform inversion.

1 INTRODUCTION

High-resolution imaging of the subsurface and detailed characterization of the geological conditions are essential for understanding the environmental changes at near-surface resulting from the exploitation of human needs and for assessing the environmental risks associated with such changes. In this context, geophysical methods play an important role in understanding this complex geological evolution by continuously monitoring the physical properties of the subsurface. Of all geophysical techniques, ground-penetrating radar (GPR) is one of the most popular approaches for obtaining high-resolution images of the near-surface because of its full use of high-frequency electromagnetic waves. While rough estimates of target locations can be obtained by directly access to GPR data profiles, however, appropriate pro-

cessing is required to obtain high-resolution images and accurate dielectric permittivity and electric conductivity of the subsurface.

To accurately characterize the physical properties of the subsurface, several approaches have been proposed based on different physical assumptions by minimizing the discrepancy in some aspects between the acquired GPR profiles and the simulated information using the given model. Ray-based tomography is one of the most popular inversion methods and is widely used to invert the physical properties of the subsurface by using traveltime information from GPR data due to its high-computational efficiency (Holliger *et al.* 2001; Busch *et al.* 2012; Deiana *et al.* 2018). However, the resolution of inversion using tomography is very limited as only a small proportion of the acquired GPR signal information is utilized.

In contrast to tomography-based methods, full waveform inversion (FWI) has attracted much attention in recent years due to its ability to fully exploit a maximum amount of wavefield information in acquired signal profiles, including amplitude, traveltime, waveform, and phase (Virieux & Operto 2009). By comparing the misfits of all aspects of a signal between recorded and simulated data, FWI is sensitive to all structural and physical properties of the subsurface that may have an impact on wave propagation, which greatly improves the spatial resolution of FWI's reconstructions of subsurface structures and physical parameters. The pioneering work on FWI was first proposed by Tarantola (1984) for solving seismic inversion problems. Since then, FWI has been rapidly developed and enhanced to accommodate different physical assumptions for complex media, in both the time and frequency domains (Ernst, *et al.* 2007a; Ernst *et al.* 2007b; Lavoué *et al.* 2014; Feng *et al.* 2019).

For GPR, FWI attempts to reconstruct the relative dielectric permittivity ϵ_r and the electric conductivity σ , which can be easily adapted to different acquisition settings, for example, surface GPR (Busch *et al.* 2012; Feng *et al.* 2019; Wang & Feng 2020), off-ground GPR (Minet *et al.* 2009; Kalogeropoulos *et al.* 2013), borehole GPR (Gloaguen *et al.* 2007; Yu *et al.* 2022), or crosshole GPR (Ernst, *et al.* 2007a; Giannakis *et al.* 2021). Despite all the progress that has been made, there are still many issues that need to be addressed for FWI of GPR data. For instance, due to the limited illumination caused by the local acquisition setting, FWI of surface-recorded GPR profiles is usually treated as an ill-posed inversion problem. The complexity of the wave propagation equation also leads to the fact that FWI is an optimization problem with a high degree of non-linearity. This makes FWI computationally intensive and difficult to obtain the optimal solution in a direct way (Plessix 2006). The general approach to solving FWI is the adjoint state method, where the gradient of the quantity of interest, that is a form of data discrepancy, with respect to the physical parameters is computed using the dual form of the optimization constraints, that is the physical laws of wave propagation (Plessix 2006; Yedlin 2010). Although the adjoint state method allows FWI to compute the gradient efficiently to a certain extent, the optimization of FWI still relies heavily on the convexity assumption, which makes it suffer severely from local minima trapping and the cycle skipping problem. In other words, a good starting point is required for FWI to converge to the global minimum, that is the desired physical parameters.

A common practice is to use other methods, such as traveltime-based tomography, to provide a good initial model for FWI and thus avoid FWI optimization falling into local minima traps (Virieux & Operto 2009). However, in realistic and complex geological environments, models obtained by tomographic methods may not be sufficient as starting models for FWI, which will be discussed later in this paper. An alternative is to perform the multiscale FWI, where the low-frequency information is first extracted from acquired profiles to invert the low-wavenumber model, which is then used as the initial model in the high-frequency inversion (Meles *et al.* 2010; Bunks *et al.* 2012). The multiscale FWI is robust, but it still demands a relative high-quality initial model, and its inversion results heavily depend on the frequency-band selection strategy and the presence of low-frequency information in the data. The former requires substantial manual intervention and trial-and-error testing, while the latter may be absent in real data due to limitations in data acquisition equipment. Many studies have been conducted to artificially mimic the low-frequency content of recorded data by reformulating the objective of FWI, for instance, the envelope-based

objectives (Liu *et al.* 2022), the deconvolution-based objectives (Feng *et al.* 2023), and the optimal transport distances (Métivier *et al.* 2016). These objective reformulation methods are relatively robust, but still cannot adapt to some realistic situations, such as complex data containing large amounts of noise or multiple arrivals, especially in the absence of good initial models (Pladys *et al.* 2021).

Unlike these physics-deterministic inversion methods, where the subsurface parameters are grid-based and usually governed by physical laws, deep learning, on the other hand, has become one of the most powerful tools in various scientific and engineering fields due to its data-driven nature and powerful representation of big data. For example, data-driven deep learning is now widely used in the field of geophysical data processing, such as first-arrival picking (Guo *et al.* 2020; Yuan *et al.* 2020), noise attenuation (Sun *et al.* 2020b; Yang *et al.* 2021), and data interpolation (Jia *et al.* 2018). However, for complex tasks, such as geophysical imaging and inversion, studies have shown that the performance of data-driven deep learning is severely hampered by a lack of robustness and generalization ability (Sun *et al.* 2021). A simple but promising direction is to build physics-guided deep learning inversions in which deep neural networks are (jointly) constrained by the physical laws (Sun *et al.* 2021, 2020a; Zhang *et al.* 2020). The physics-guided strategy can greatly enhance the robustness of deep learning models and improve their generalization ability to a certain extent, but still faces major challenges in imaging complex geological formations. Therefore, how to integrate deep learning with known physical knowledge to address the existing challenges of traditional inversion methods is a pressing issue.

In deep learning optimization, research has shown that deep neural networks tend to learn low-frequency information before learning high-frequency content, which is known as the frequency principle (or frequency/spectral bias) (Rahaman *et al.* 2019; Basri *et al.* 2020). There are similarities between the frequency principle phenomenon of deep learning optimization and multiscale FWI in that they both favour the use of low-frequency information over high-frequency information. Therefore, it is possible to integrate FWI with deep learning using the frequency principle to reduce the problem of FWI's strong dependence on the initial model. The seismic implicit full waveform inversion (IFWI) implemented by Sun *et al.* (2023) is the first attempt of this idea, which has recently been further improved (Du *et al.* 2023) and extended to elastic cases (Zhang *et al.* 2023).

In this paper, we extend the application of IFWI from seismic to GPR data, with a specific focus on the challenges posed by the sensitivity to initial models. Due to the inherent discrepancies in parameter sensitivities, GPR multiparameter FWI often necessitates weighting factors to balance parameter contributions. However, our work departs from this strategy by simultaneously inverting permittivity and conductivity without the use of any weighting factors, demonstrating the robustness of IFWI in this context. Furthermore, we present a comprehensive analysis of how hyperparameter settings influence IFWI performance on GPR data, culminating in practical guidelines for optimizing these settings. Specifically, we will first briefly review the FWI formulation for GPR and then introduce the concepts of implicit neural representations and frequency principle in deep learning optimization. After that, we discuss how to reformulate FWI as a physics-constrained implicit neural representation problem, where FWI can be driven by the frequency principle of deep learning optimization. Moreover, we analyse the hyperparameters selection of network design and its performance of IFWI will be examined by numerical examples.

2 THEORY

2.1 Traditional FWI formulation for GPR

The electromagnetic wave propagation can be delineated by the Maxwell's equations, which can be succinctly expressed in the following compact form:

$$\begin{bmatrix} -\varepsilon(\mathbf{m})\partial_t^2 - \sigma(\mathbf{m})\nabla \times \\ \nabla \times \end{bmatrix} \begin{bmatrix} \mathbf{E}(\mathbf{m}, t) \\ \mathbf{H}(\mathbf{m}, t) \end{bmatrix} = \begin{bmatrix} \mathbf{J}^s(\mathbf{m}, t) \\ 0 \end{bmatrix} \quad (1)$$

where, $\mathbf{E}$ and $\mathbf{H}$ denote the electric and magnetic fields in the space-time domain, respectively, which are induced by the source current density $\mathbf{J}^s$. $\mu = \mu_r \mu_0$ represents permeability, which is assumed to be constant. Here, μ_r denotes the relative permeability, and μ_0 is the permeability of free space. The permittivity and conductivity of the medium are denoted by $\varepsilon(\mathbf{m})$ and $\sigma(\mathbf{m})$, respectively, and are distributed within the model. The notation ∂_t signifies the partial derivative with respect to time, and $\nabla \times$ represents the curl operator in the spatial domain.

The eq. (1) can be solved in time domain by discretizing it into a finite difference formulation. In this paper, a second-order discretization in temporal and spatial dimensions with a convolutional perfectly matched layer boundary condition are adopted for all forward simulations. Moreover, for simplicity, we consider a non-magnetic medium where the permeability is homogeneous and equal to $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$.

The ultimate goal of the GPR inversion problem is to find a set of subsurface models from which the generated simulation best matches the local measurements. Therefore, a simple way to evaluate inverted models is to calculate the discrepancy between the simulation and locally measured data. A ℓ_2 -norm of the data misfit is commonly used as the objective function for FWI of GPR, written as

$$\mathcal{L}(\mathbf{m}) = \frac{1}{2} \sum_{s,r,t} \|\mathbf{d}(\mathbf{m}) - \mathbf{d}_{\text{obs}}\|^2, \quad (2)$$

where $\mathbf{m}$ denotes the vector of model parameters to be inverted. Instead of the permittivity ε and conductivity in S m^{-1} , the relative permittivity $\varepsilon_r = \varepsilon/\varepsilon_0$ and conductivity in mS m^{-1} are usually used for inversion when considering the stability of the inversion, i.e., $\mathbf{m} = [\varepsilon_r, \sigma * 1000]^T$. $\mathbf{d}$ represents the forward simulation using $\mathbf{m}$ with the appropriate acquisition geometry setting, and $\mathbf{d}_{\text{obs}}$ is the observed data, that is the electrical component $\mathbf{E}_y$ in the GPR acquisitions. s, r, t denote source, receiver, and time, respectively.

With the optimization of FWI, the model is updated at the k th iteration as

$$\mathbf{m}_k = \mathbf{m}_{k-1} - \alpha \Delta \mathbf{m}_k, \quad (3)$$

where, $\mathbf{m}_0$ for $k = 1$ represents the starting model for FWI, $\Delta \mathbf{m}_k$ denotes the model updates obtained at the k th iteration, which are associated with the gradient of the objective function with respect to the model parameters $\mathbf{g}_k$, α is the step-size in the optimization of FWI and is also referred to as the learning rate in deep learning optimization.

2.2 RNN-based FWI formulation for GPR

In the traditional FWI, the gradient $\mathbf{g}$ is typically computed by backpropagating the data misfit using the adjoint state methods. Sun *et al.* (2020a) points out that reformulating the forward wave propagation into the framework of recurrent neural network (RNN) enables obtaining the gradient using the automatic differentiation

with the chain rule, which can be written as:

$$\mathbf{g} = \frac{\partial \mathcal{L}}{\partial \mathbf{m}} = \sum_{s,r,t} \frac{\partial \mathcal{L}}{\partial \mathbf{d}} \frac{\partial \mathbf{d}}{\partial \mathbf{m}}. \quad (4)$$

Sun *et al.* (2020a) further demonstrated that the gradient obtained through automatic differentiation is theoretically consistent with that derived from adjoint state approaches. Details of the proven derivation can be found in Sun *et al.* (2020a). Here, we refer to FWI within an RNN framework as RNN-based FWI. With the RNN framework, FWI optimization can be treated as the training stage of a theory-designed RNN, where the model vector $\mathbf{m}$ is the trainable parameters. The RNN-based FWI formulation can also be easily adapted for electromagnetic inversion, as demonstrated by Hu *et al.* (2022).

We emphasize that the RNN-based FWI is theoretically consistent with traditional FWI, with the key distinction that the former eliminates the need for manual gradient derivation in the implementation of FWI. Another advantage of implementing FWI using an RNN framework is that the optimization process can easily be accelerated using GPU-based deep learning optimization algorithms, such as adaptive momentum (i.e. Adam; Kingma 2017).

2.3 Implicit neural representation for FWI

Although the theory-designed RNN framework allows us to perform FWI from a deep learning perspective, such an implementation still faces the same challenges as traditional FWI. To better exploit the features and benefits of deep learning, we need a deeper integration of neural networks and FWI in terms of the underlying theory. Implicit neural representation (INR) denotes a novel paradigm within the domains of computer graphics and computer vision, which seeks to represent intricate 3D shapes and scenes without explicit specification of their geometric attributes. Diverging from conventional methodologies reliant on explicit representations like meshes or voxels, INR leverages deep neural networks to learn an implicit function that can generate the desired shape or scene.

In this framework, a neural network is trained to predict a signed distance function (SDF) for each point within the 3D space. The SDF specifies the distance from a given point to the nearest surface of the object while providing information pertaining to the point's position with respect to the shape's interior or exterior. By training the neural network on a data set of 3D shapes or scenes, it learns to generalize and generate accurate SDF values for unseen points.

With this in mind, we give a general definition of INR: the use of deep neural networks to learn a continuous and implicit function to represent the distribution of features of interest $\Phi(\mathbf{x})$ within a certain domain $\mathbf{x} \in \Omega$. They can be expressed mathematically as

$$\tilde{\Phi}(\mathbf{x}) = \mathcal{N}_{\Theta}(\mathbf{x}), \quad (5)$$

where $\mathcal{N}_{\Theta}$ represents the deep neural network with trainable weight vector Θ . Here, the features of interest can be given directly, which leads to data-driven INR tasks, or they can be subject to the laws of physics, in which case INR falls under the category of physics-guided deep learning problems. Specifically, a data-driven INR is optimized by minimizing the discrepancy between the given features and the network's output, that is $\|\Phi - \mathcal{N}_{\Theta}(\mathbf{x})\|^2$. On the other hand, the loss function of a physics-guided INR problem is $\|\mathcal{C}(\mathbf{x}, \mathcal{N}_{\Theta}(\mathbf{x}))\|^2$, where $\mathcal{C}$ denotes the physical laws obeyed by features of interest.

The inversion of GPR data can also be fitted into the framework of physics-guided INR, in which the features of interest Φ is the

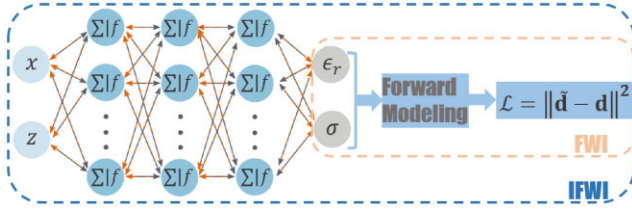


Figure 1. The schematic workflow of implicit full waveform inversion, where Σ and f denote the linear transformation and the activation function in the calculation of a single neuron.

model parameters $\mathbf{m}$ to be inverted. In the case of GPR FWI, the physics constraint $\mathcal{C}$ is determined by data discrepancy between the observation and the simulated data, which is associated with the Maxwell's equations. Mathematically, reformulating FWI into the physics-guided INR framework can be expressed as

$$\mathcal{L}(\Theta) = \frac{1}{2} \sum_{s,r,t} \|\mathcal{F}(\mathcal{N}_{\Theta}(\mathbf{x})) - \mathbf{d}_{\text{obs}}\|^2, \quad (6)$$

where $\mathbf{x}$ represents the spatial coordinates of the inverted region, $\mathcal{N}_{\Theta}$ is the representation neural network with spatial coordinates as input, and its expected output is the desired model parameters, that is $\mathbf{m}(\mathbf{x}) \approx \mathcal{N}_{\Theta}(\mathbf{x})$. $\mathcal{F}$ represents the forward modelling operator for Maxwell's equations. In the following, we refer to the INR form of FWI as implicit full waveform inversion, or IFWI for short. The original concept of IFWI was first introduced by Sun *et al.* (2023), where the subsurface velocity model was inverted using IFWI with seismic data. However, due to the different characteristics of electromagnetic wave propagation, the applicability and performance of the IFWI using GPR profiles needs to be further investigated, which is the main purpose of this paper. The schematic workflow of IFWI using GPR profiles is shown in Fig. 1.

Unlike conventional or RNN-based FWI that relies on grid-based subsurface representations, IFWI utilizes a single neural network to represent all physical parameters, where this representation is an implicit and continuous function. One of the key advantages of implicit and continuous representation is its ability to handle complex and detailed subsurface structures without the need for explicit parameterization or discretization, as long as physical constraints can be applied in a mesh-free manner. This allows greater flexibility and efficiency in representing a wide range of physical parameters, while the coupling effects between different parameters (i.e. crosstalk) can be implicitly represented by the neural network. Additionally, the continuous representation can naturally handle the case where the same model parameters are represented at different scales, for example in multiphysics inversions where the same model parameters at different scales are governed by different physical laws and observations. Moreover, reformulating FWI as a physics-guided form of INR also allows us to implement FWI in a deep learning optimization approach facilitated by the frequency principle.

2.4 Frequency principle

The frequency principle of deep learning optimization is a theoretical framework that seeks to provide insights into the behaviour and dynamics of deep neural networks during the learning process. According to the frequency principle, low-frequency information refers to the coarse, global, or low-resolution aspects of the data, whereas high-frequency information pertains to fine-grained, local, or high-resolution details. In the early stages of training, neural

networks exhibit a preference for capturing the low-frequency components of the data, ensuring that networks can first gain a broad understanding of the data distribution (i.e. developing a coarse approximation of the underlying patterns) and guide subsequent learning processes. As training progresses, the networks gradually refine their models and allocate resources towards capturing the higher-frequency details, which correspond to more intricate and localized features within the data.

Fig. 2 shows the frequency principle phenomenon using a numerical experiment, where the signal to be learned is indicated as blue and the network outputs at different training epochs are shown as red. From top to bottom, comparisons between the truth signal and the result learned by the network at training epochs [0, 1000, 4000, 9000] are delineated in both time and frequency domains in the left and right columns of Fig. 2, respectively. Both the time and frequency domain comparisons reflect the fact that neural networks prefer to learn low-frequency information before learning high-frequency content of the target data. In addition, in the early training phase, the neural network is able to learn the low-frequency information of the target data while maintaining its stability at high frequencies so that it does not deviate from the high-frequency information in the target data. Therefore, using the frequency principle, IFWI can first automatically reconstruct the low-frequency component of the subsurface models, and then reconstruct the high-frequency details based on the low-frequency component. In other words, IFWI is less dependent on the initial model, and therefore has more flexibility.

3 NUMERICAL EXAMPLES

To verify our inference that IFWI is less dependent on initial models, in this section we investigate the performance of IFWI with poor initial models. Two synthetic models, the Cross-shape and the Overthrust models, are used to generate GPR profiles, which are treated as observations during the inversion. Further comparisons between FWI and IFWI are also discussed in this section to provide an intuitive insight into IFWI. It is worth noting that all FWI experiments are conducted within an RNN-based framework to serve as baselines for fair comparison with IFWI, as RNN-based FWI provides greater flexibility in selecting deep learning optimization algorithms, required for IFWI. In addition, all numerical experiments of FWI and IFWI in this section are performed in the time domain with simultaneous multiparameter inversion using a gradient-based Adaptive momentum (Adam; Kingma 2017) optimization algorithm due to its fast convergence rate. The detailed comparisons of Adam and conventional optimization algorithms, including conjugate gradient and Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS), can be found in Sun *et al.* (2020a).

3.1 Cross-shape model

Before examining the performance of FWI and IFWI, it is necessary to investigate the behaviour of multiparameter GPR inversion using full waveforms. A homogeneous model with two Cross-shaped anomalies, initially introduced by Meles *et al.* (2011), is used in the proposed study, as shown in Fig. 3. Each model contains 10 201 grid points with a shape of 101×101 and a grid spacing of 10 cm. The relative permittivity and conductivity of the homogeneous background are $\epsilon_r = 4$ and $\sigma = 3 \text{ mS m}^{-1}$, respectively. The anomaly located in the upper-left region has a relative permittivity $\epsilon_r = 1$

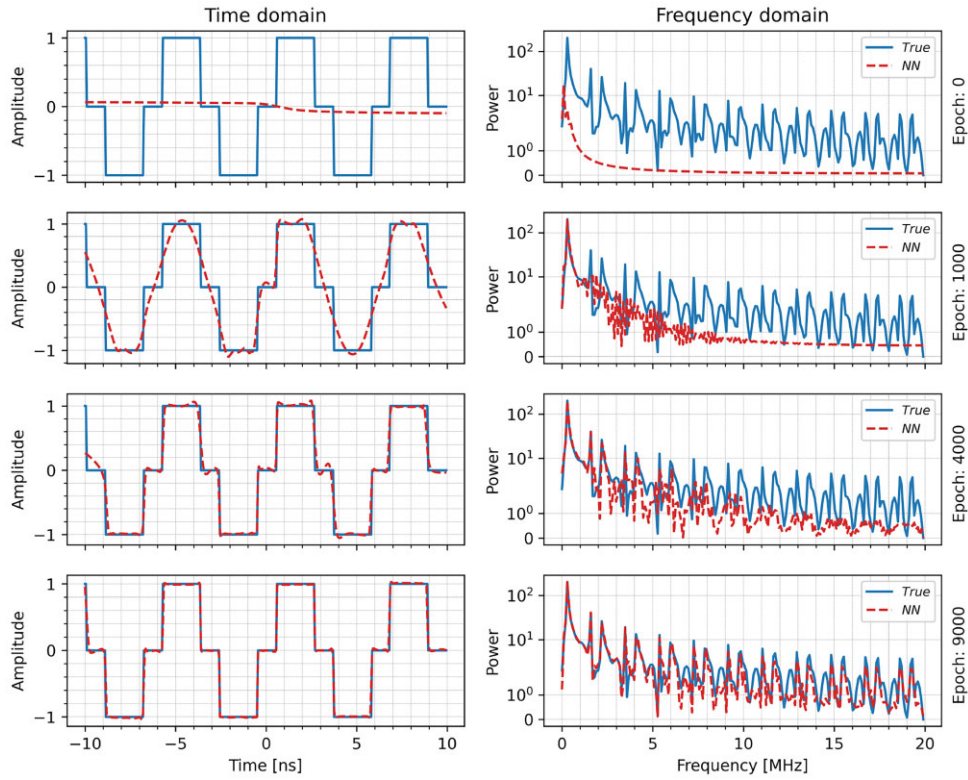


Figure 2. The phenomenon of frequency principle in deep learning optimization; the solid line indicates the signal to be learned, while the dashed line represents the results learned by the neural network at different epochs, from top to bottom, the first to the 9000th epoch; left and right columns denote the delineations in time and frequency domains, respectively.

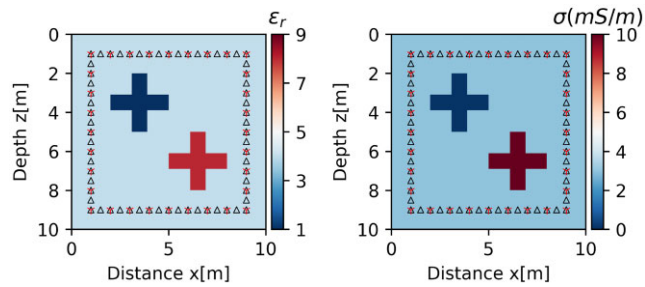


Figure 3. The Cross-shape models and corresponding acquisition settings: (a) the relative permittivity and (b) the conductivity. The receiver locations are marked with triangles and source locations are delineated as the cross lines. Note that we assume the antennas to be perpendicular to the plane of observation (TE mode) following Lavoué *et al.* (2014), whereas Meles *et al.* (2011) use in-plane antennas (TM mode).

and $\sigma = 0.1 \text{ mS m}^{-1}$, and the physical parameters for the lower-right anomaly are $\epsilon_r = 8$ and $\sigma = 10 \text{ mS m}^{-1}$.

The Cross-shape model presents practical challenges when incorporating in-plane antennas operating in the transverse magnetic (i.e. TM) mode and broadside antennas that are perpendicular to the plane of observation, operating in the transverse electric (i.e. TE) mode, due to the presence of transmitting and receiving antennas within the model. While the TM mode is relatively more appropriate for such ‘borehole’ acquisition setups (Meles *et al.* 2011), the TE mode can also be used for synthetically validating the effectiveness of proposed algorithms, as demonstrated in Lavoué *et al.* (2014). Considering that the primary goal of this work is high-precision GPR inversion with surface acquisitions, we adopt the

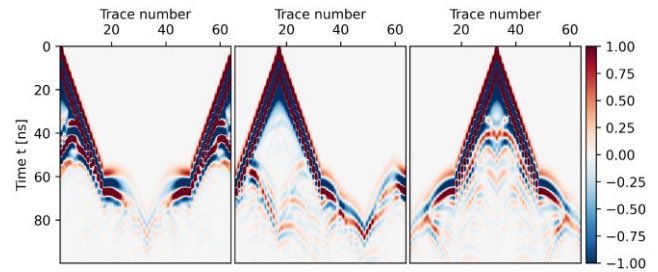


Figure 4. Time-domain plots of three representative shot gathers generated using the truth models shown in Fig. 3.

TE mode for the Cross-shape model to ensure consistency with our later experiments, while conducting theoretical validation of IFWI’s effectiveness on a simple subsurface model with comprehensive acquisitions.

As shown in Fig. 3, these anomalies are surrounded by 32 transmitting and 64 receiving antennas in a perfect illumination configuration, where locations of receivers and sources are indicated as black triangles and red cross lines, respectively. For synthetic GPR data simulation, a Ricker wavelet with a single dominant frequency of 100 MHz is chosen as the source in this paper. Fig. 4 shows three representative shot gathers in the time domain, where these three sources are located in the top-left corner, top-right corner and bottom-right corner, respectively. The trace numbers of the receiving antennas are 1 to 64, in the order of top-left, top-right, bottom-right, bottom-left and back to top-left corners. The maximum recording time is 100 ns with an interval of 0.2 ns.

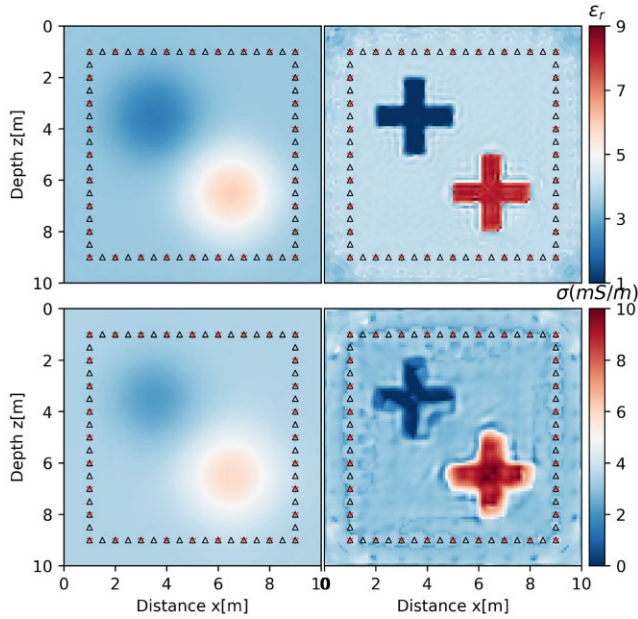


Figure 5. Inversion results of FWI using Adam with smooth initial models. Left column: initial models and right column: FWI results. Top row: the relative permittivity and bottom row: the conductivity.

To illustrate the dependence of FWI on the initial model more intuitively, we first perform FWI on the GPR profile of a given initial model, as shown in the left column of Fig. 5. In initial models of relative permittivity and conductivity, the locations of both anomalies can be observed, although their exact shape remains unknown. The right column of Fig. 5 shows the final inversion results of FWI using these smoothed initials. As shown in the upper-right corner of Fig. 5, FWI is able to accurately reconstruct the size and shape of the anomalies at correct locations, as well as its exact values in terms of relative permittivity. In terms of conductivity, FWI successfully reproduces the location and size of the anomalies with near exact shape, although small background distortions are observed. Fig. 6 depicts a comparison of data for the source located in the right-bottom corner. Specifically, Figs 6(a)–(c) respectively display the data simulated using ground-truth of the Cross-shape models, smooth initials, and FWI models as shown in Fig. 5. The data residuals between the ground-truth and smooth initials are illustrated in Fig. 6(d), while Fig. 6(e) shows the data residuals between the ground-truth and FWI model in the right column of Fig. 5. Therefore, it can be said that with a good initial model, FWI has the ability to provide detailed information of subsurface anomalies by minimizing the data misfits of GPR observation and simulated data without encountering local minimum traps.

However, building starting models that closely resemble reality for FWI proves to be an immensely challenging task, particularly during the initial stages of geophysical prospecting, where prior information is lacking. Hence, it becomes imperative to assess the performance of FWI using low-quality initial models. Figs 7(a) and (b) show homogeneous models of relative permittivity $\epsilon_r = 4$ and conductivity $\sigma = 3 \text{ mS m}^{-1}$. For the same source located in the right-bottom corner, data comparisons between the ground-truth and homogeneous models are demonstrated in Figs 8(a) and (b). Starting with the homogeneous background models, the final FWI results of relative permittivity and conductivity are plotted in Figs 7(c) and (d), respectively. The results show that, starting from homogeneous models, FWI can only provide an approximate location and

fuzzy shape of the relative permittivity, while large distortions appear in the inversion results of the conductivity. Fig. 8(c) displays the simulated data using these inverted models, while large residuals between this simulated data and the ground-truth observations are observed in Fig. 8(d). We emphasize that, unlike conventional strategies requiring a weighting factor for conductivity – where the selection of an appropriate factor heavily relies on the availability of prior information or extensive trial-and-error experiments Lavoué *et al.* (2014) – all experiments in this paper use only the approximate global mean and standard deviation, without incorporating any additional weighting factors. Therefore, it is possible that better inversion results may be achieved with an appropriate weighting factor. However, the result of this experiment are sufficient to indicate that traditional FWI is strongly influenced by the local minima trap, leading to a strong reliance on the initial model.

Studies have shown that multiscale strategy has the capacity to mitigate, to some extent, FWI's dependence on initial models (Meles *et al.* 2011; Bunks *et al.* 2012). For comparison, the multiscale FWI is also implemented in the time domain using dominant frequencies of [60, 80, 100] MHz, respectively, in three sequential stages. The corresponding 'recorded' GPR data are obtained using the Wiener filter (Feng *et al.* 2019). The final results of multiscale FWI are delineated in Figs 7(e) and (f), respectively. Comparing the one-step FWI, it becomes evident that multiscale FWI excels in accurately reconstructing Cross-shaped anomalies in relative permittivity. In terms of conductivity, multiscale FWI offers an approximate representations of two anomalies, albeit at a reduced level of resolution. That is particularly notable in background distortions, which can be mitigated by adding a regularization term, such as the total variation regularization. Regarding the data comparison, Fig. 8(e) reveals that the data simulated using multiscale FWI models exhibit a closer alignment with the observations in Fig. 8(a). This is further supported by the smaller data residuals depicted in Fig. 8(f), in comparison to FWI. There is potential for enhanced multiscale FWI results through the addition of a weighting factor and alternative frequency band selections, however, the experimental results are sufficient to show that by initiating with a lower frequency band, multiscale FWI can mitigate the local minima of FWI's objective and reduce the starting model's dependence to a certain extent.

Next, we perform the GPR inversion using the proposed method, that is IFWI. A neural network is first required to implicitly represent the subsurface parameters, that is in this case the relative permittivity and conductivity. As shown in Fig. 1, this representation network takes spatial coordinates (depth and distance) as input and outputs the reconstructed physical parameters. For the network architecture design, we choose a simple multilayer perceptron (MLP) with four hidden layers and 128 neurons per layer. Each hidden layer consists of a linear transformation $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and a nonlinear activation function. Instead of the rectified linear unit (ReLU), we select a scaled periodic sinusoidal function $\mathbf{y} = \sin(\omega_0 \mathbf{x})$ as the activation function to increase the nonlinearity of the representation network and its ability to represent high-frequency content. Here, similar to (Sitzmann *et al.* 2020; Sun *et al.* 2023), a fixed scaling factor $\omega_0 = 30$ is chosen in this experiment. Considering that the weights of the network are usually initialized by randomly drawing from a normal distribution with zero mean and small standard deviation, a zero-mean network output may greatly speed up the convergence process. Therefore, instead of outputting the relative permittivity and conductivity ($\mathbf{m}$) directly, their scaling $\tilde{\mathbf{m}}$ by the global mean μ and standard deviation ζ is considered as the representation network's output, that is $\mathbf{m} = \tilde{\mathbf{m}}\zeta + \mu$. Note that the global mean and

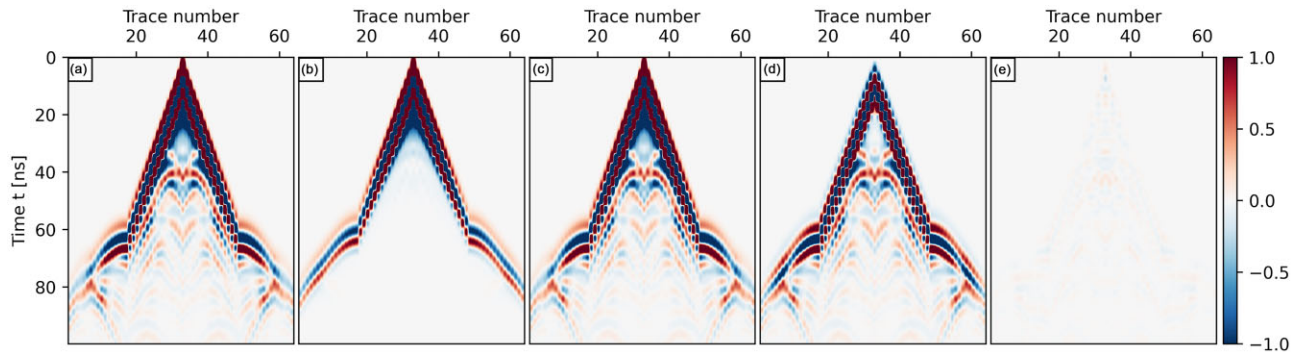


Figure 6. Data comparisons with the source located in the bottom-right corner using (a) the ground-truth models in Fig. 3, (b) smooth initials and (c) inverted models in Fig. 5. (d) and (e) denote residuals (a), (b) and (a)–(c), respectively.

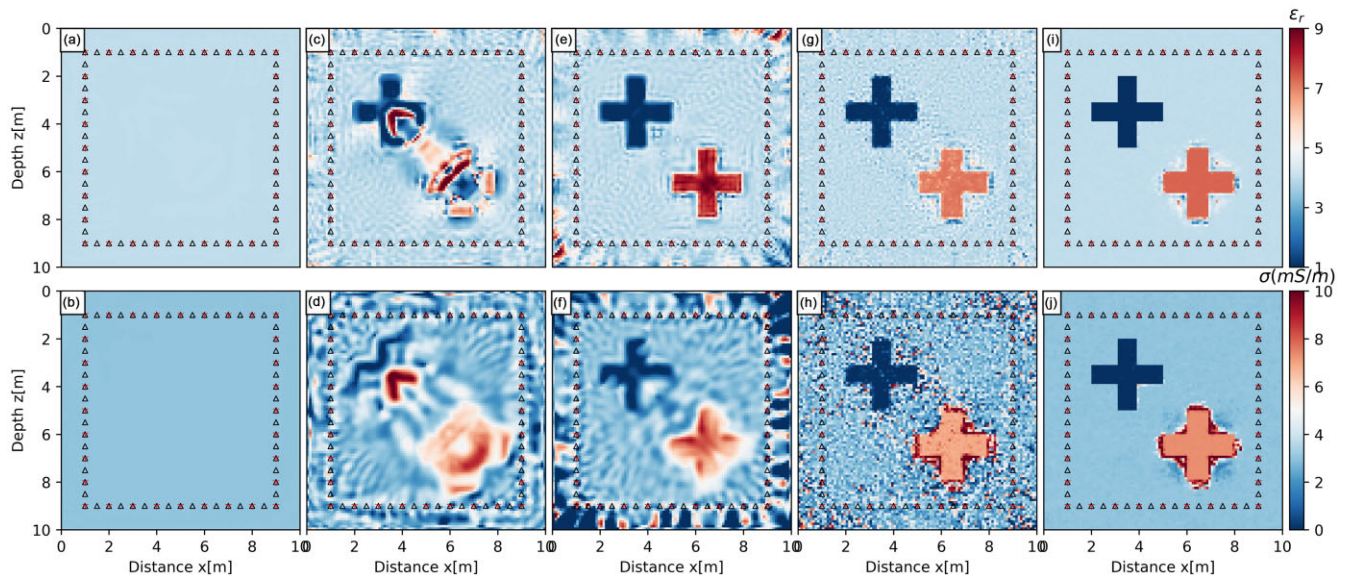


Figure 7. Comparisons of FWI, multiscale FWI and IFWI. Top row: relative permittivity and bottom row: conductivity. (a) and (b): homogeneous initial models; (c) and (d): FWI; (e) and (f): multiscale FWI; (g) and (h): IFWI; (i) and (j): IFWI with dropout regularization.

standard deviation do not need to be perfectly accurate, but are used only to ensure that the output of the network lies on a normal distribution with an approximate zero mean and variance of one, thus accelerating the convergence of IFWI. Such global means and standard deviations can be calculated using information from boreholes in the target area or can be roughly estimated based on knowledge of the target area.

The representation network is first pretrained on the homogeneous backgrounds depicted in Figs 7(a) and (b), ensuring that IFWI employs the same initial models as FWI and multiscale FWI. The sole distinction between FWI and IFWI lies in the fact that, unlike traditional FWI where the subsurface model parameters $\mathbf{m}$ are updated, IFWI exclusively updates the weights of the representation network. The final results of IFWI starting with homogeneous models are illustrated in Figs 7(g) and (h), respectively. Compared to FWI (Figs 7c and d), IFWI excels in avoiding local minima and accurately reconstructing the size and shape of anomalies in their correct locations, similar to multiscale FWI. Notably, no manual frequency band selections are involved in the implementation of IFWI. Additionally, without a scaling factor, the conductivity inversion results from IFWI exhibit a slightly more precise shape than

those from multiscale FWI (Figs 7e and f), particularly evident in Figs 8(g) and (h), where a closer examination of the simulated data using these IFWI models reveals smaller data residuals. This preliminary result shows that by incorporating the frequency principle of deep learning optimization, IFWI can significantly reduce the dependence on the initial model and the risk of falling into local minima.

A small drawback observed in IFWI's results (Figs 7g and h) is the background distortion, especially in the conductivity, which is likely caused by the oversensitivity of the sinusoidal function with a large ω_0 to high-frequency components. Fortunately, this artifact can be eliminated by adjusting activation functions or by adding regularization to the weights or outputs of the network. The former may include reducing the ω_0 for the **sin** activation or using *tanh* and other variants of ReLU as activation, while the latter includes batch normalization or dropout for the weights and ℓ_1 –or ℓ_2 –norm for the outputs.

For this straightforward inversion setup, we suggest incorporating dropout neurons as a form of regularization, given their simplicity and ease of implementation. Figs 7(i) and (j) show the final results of IFWI using dropout regularization (default: $p = 0.2$), that is

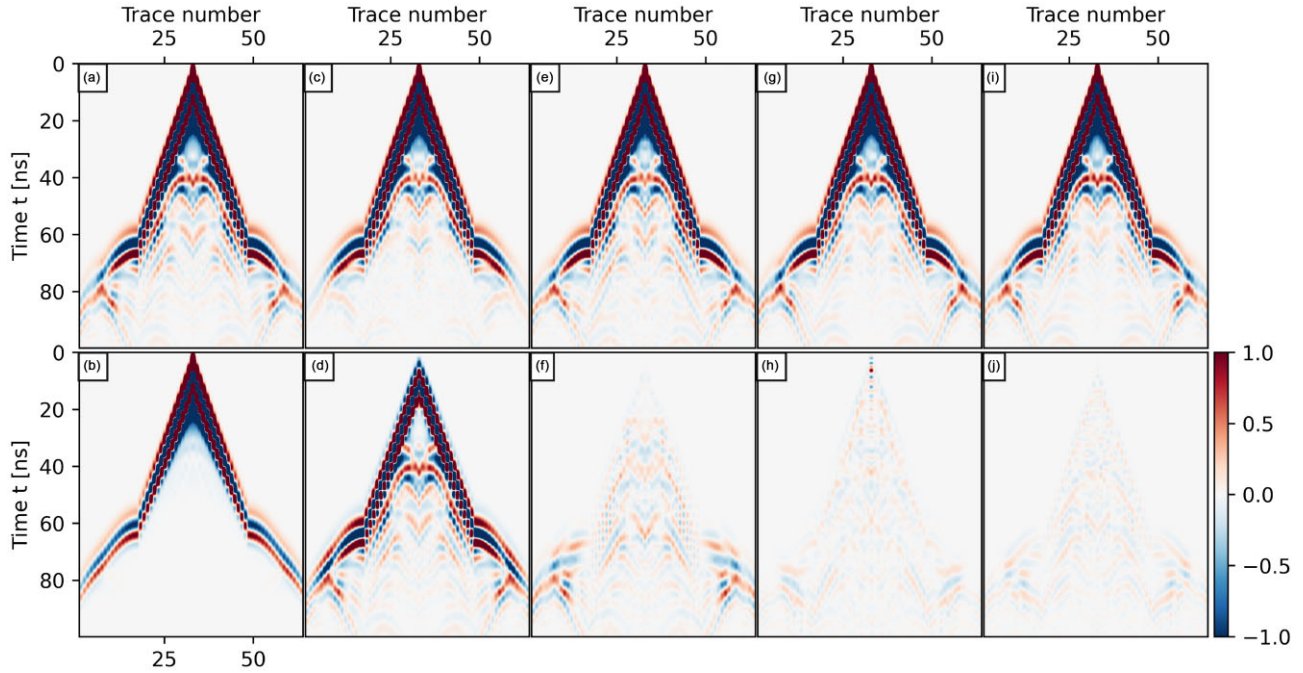


Figure 8. Data comparisons with the source located in the bottom-right corner using (a) the ground-truth; (b) homogeneous initial models in Figs 7(a) and (b); (c) FWI models in Figs 7(c) and (d); (d) residuals of (a) and (c); (e) multiscale FWI models in Figs 7(e) and (f); (f) residuals of (a) and (e); (g) IFWI models in Figs 7(g) and (h); (g) residuals of (a) and (g); (i) dropout-IFWI models in Figs 7(i) and (j); (j) residuals of (a) and (i).

randomly dropping 20 per cent of the neurons in the hidden layers during inversion. The results show that the ideal anomaly distributions with sharp boundaries and exact parameter values can be achieved using dropout-IFWI starting from homogeneous initial models. The data simulation using these models and their associated residuals are plotted in Figs. 8(i) and (j), respectively. It should be noted that the neural network design chosen in this experiment is not optimal, and better performance may be obtained with other network architectures or hyperparameter settings.

3.2 Overthrust model

We now present a more realistic benchmark of complex subsurface imaging using multioffset on-ground GPR acquisition settings. A 2-D section of the Overthrust model with a depth of 4.7 m and a width of 12.5 m is selected for the realistic configuration, where the grid size of the zone of interest is 94×250 with 5 cm grid spacing. The original Overthrust model has been appropriately adapted, resulting in a revised range of relative permittivity from 1 to 30 and conductivity from 0 to 20 mS m^{-1} , as depicted in Fig. 9. The adjusted model consists of dry sand in the shallow zone, complex fault structures in the horizontal direction from 4 to 11 m, strong attenuation layers at about 3 m depth and groundwater at the bottom. In addition, a 50 cm layer of air is added to the top of the model to simulate realistic collection, with 12 sources evenly distributed at 1 m intervals at 5 cm above the ground. The receiving antenna is located on each grid point at 5 cm from the ground. The locations of sources and receivers are indicated as white triangles and the dashed line in Fig. 9, respectively. Note that, transmitting and receiving antennas are assumed to be perpendicular to the plane of observation (TE mode), thus only the electric field is employed for inversion. The GPR profiles are collected with the above acquisition

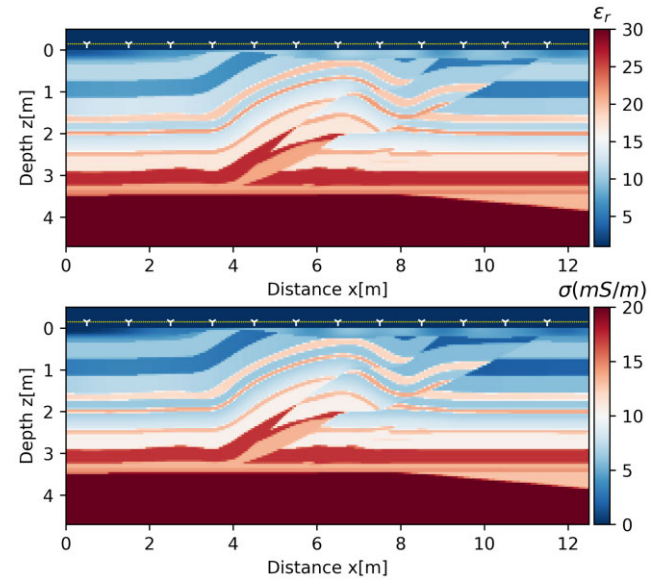


Figure 9. The 2D Overthrust model. Top: distribution of the relatively permittivity and bottom: the distribution of the conductivity.

geometry using a Ricker wavelet of 100 MHz dominant frequency, and the maximum recording time is 100 ns.

3.2.1 The multiscale FWI

To better demonstrate IFWI's advantages in handling complex subsurface structures, we first evaluate the performance of standard multiscale FWI with different starting models for comparison. The multiscale FWI of the 2D Overthrust models are also performed in

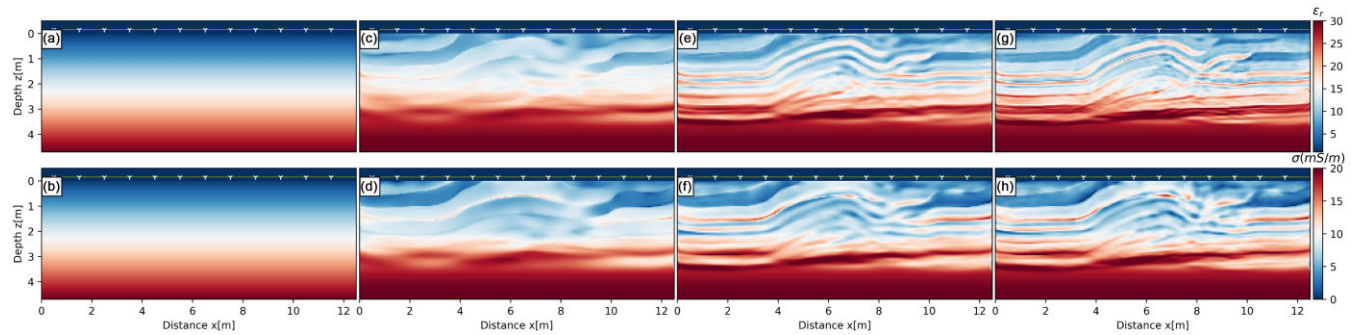


Figure 10. Multiscale FWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row. (a) and (b): accurate 1D initial models with $\epsilon_r = 1$ to 30 and $\sigma = 0$ to 20 mS m⁻¹; (c) and (d): multiscale FWI in the stage of 30 MHz; (e) and (f): multiscale FWI in the stage of 60 MHz; (g) and (h): multiscale FWI in the stage of 100 MHz.

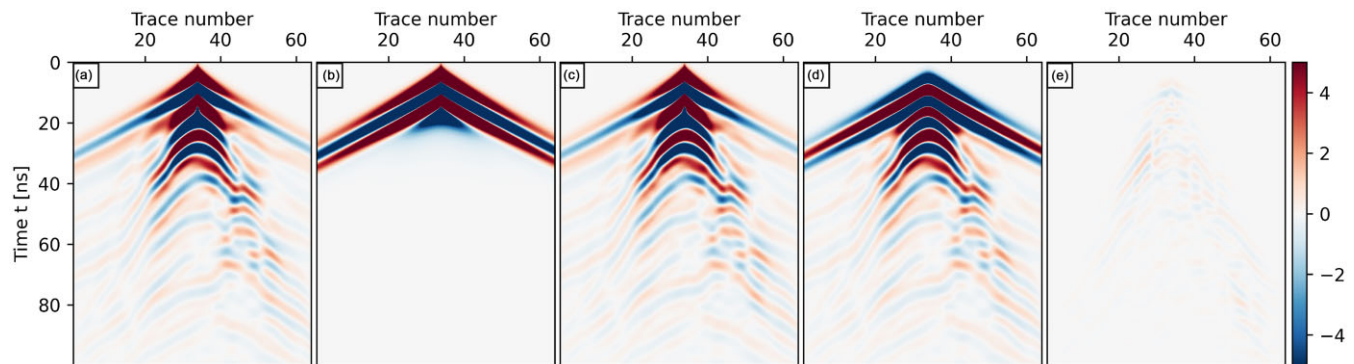


Figure 11. Data comparisons with the source located at $x_s = 6.5$ m using (a) the ground-truth Overthrust models, (b) accurate 1-D initials in Figs 10(a) and (b), (c) final multiscale FWI models in Figs 10(g) and (h). (d) and (e) denote residuals (a), (b) and (a)–(c), respectively.

the time domain by sequential stages using dominant frequencies of [30, 60, 100] MHz, respectively. Fig. 10 shows the multiscale FWI results of the Overthrust models starting from 1D initials accurately spanning the true model's range. That is to say, in Figs 10(a) and (b), the initial models for relative permittivity and conductivity are linearly increased from 1 to 30 and 0 to 20, respectively. These initial models are later referenced as accurate 1D initial models in the subsequent context. As delineated in Figs 10(c) and (d), commencing with a low-dominant frequency of 30 MHz, multiscale FWI successfully recovers the large-scale structural information in both relative permittivity and conductivity, which potentially offers high-quality starting models for the subsequent inversion stage. Figs 10(e) and (f) display the results of multiscale FWI using a dominant frequency of 60 MHz. Results demonstrate that, by introducing middle-frequency information, multiscale FWI is able to refine the inverted structures at the intermediate scale. In Figs 10(g) and (h), the small-scale structures and values of relative permittivity and conductivity are further improved by multiscale FWI with a dominant frequency of 100 MHz. Fig. 11 depicts data comparison for the source located at $x_s = 6.5$ m using the ground-truth Overthrust models, accurate 1-D initials (Figs 10a and b), and final results of multiscale FWI (Figs 10g and h). This data comparison indicates that with appropriate quality of initial models, multiscale FWI can markedly reduce the majority of data residuals and yield satisfying reconstructions of subsurface models, particularly in relative permittivity. However, there remains a small fraction of residuals near the offset, primarily due to inaccuracies in the inversion of the complex fault area in conductivity, as shown in Fig. 10(h).

Next, we make slight adjustments to the 1-D initial models, reducing the maximum values of relative permittivity from 30 to 25 and conductivity from 20 to 15 mS m⁻¹. The modified 1D initial models are later referred to as inaccurate 1D initial models and are plotted in Figs 12(a) and (b). With the same inversion strategy settings, we perform multiscale FWI on overthrust models, beginning with the inaccurate 1D initial models. The inversion results, obtained using dominant frequencies of [30, 60, 100] MHz, are presented in Figs 12(c)–(h), respectively. As shown in Figs 12(c) and (d), with the inaccurate 1-D initials, the multiscale FWI using a low-dominant frequency of 30 MHz fails to accurately reconstruct the variations at the global scale, particularly in the horizontal direction. While the inversion results show slight improvement when high-frequency information is introduced in subsequent inversion stages (Figs 12e–h), multiscale FWI with sequential stages of [30, 60, 100] MHz starting from the inaccurate 1-D initials still falls short of accurately representing the complexity of the 2-D Overthrust models. The data comparison for the source located at $x_s = 6.5$ m between ground-truth, inaccurate 1D initials, and final multiscale FWI models is depicted in Fig. 13. In comparison to Fig. 11(b), it is evident that Fig. 13(b) exhibits less resemblance with the ground-truth shot. This discrepancy is further highlighted by the larger residuals illustrated in Fig. 13(d), indicating the less accuracy of the initial models. Consequently, the final results of multiscale FWI with inaccurate 1-D initial models still exhibit significant data residuals as depicted in Fig. 13(e). Thus, it is reasonable to conclude that while multiscale FWI exhibits less dependence compared to conventional FWI, it still relies somewhat on initial models.

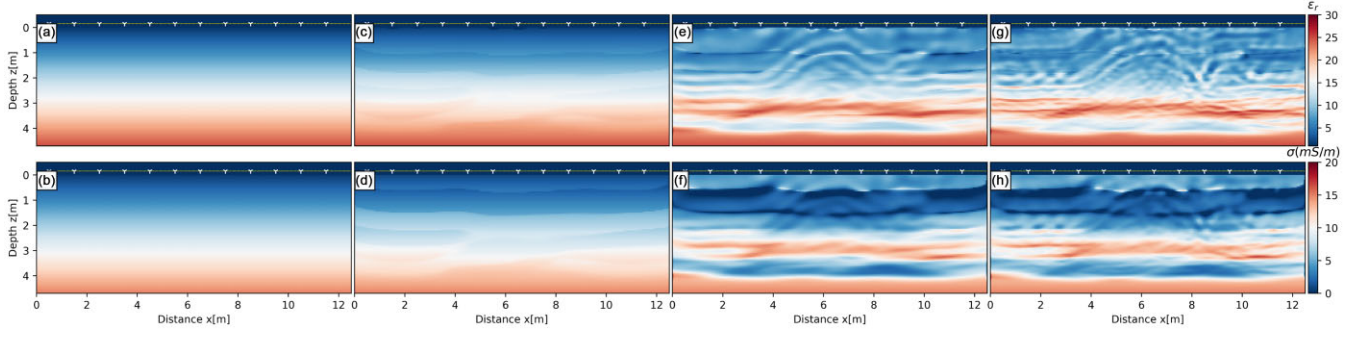


Figure 12. Multiscale FWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row. (a) and (b): inaccurate 1-D initial models with $\epsilon_r = 1$ to 25 and $\sigma = 0$ to 15 mS m^{-1} ; (c) and (d): multiscale FWI in the stage of 30 MHz; (e) and (f): multiscale FWI in the stage of 60 MHz; (g) and (h): multiscale FWI in the stage of 100 MHz.

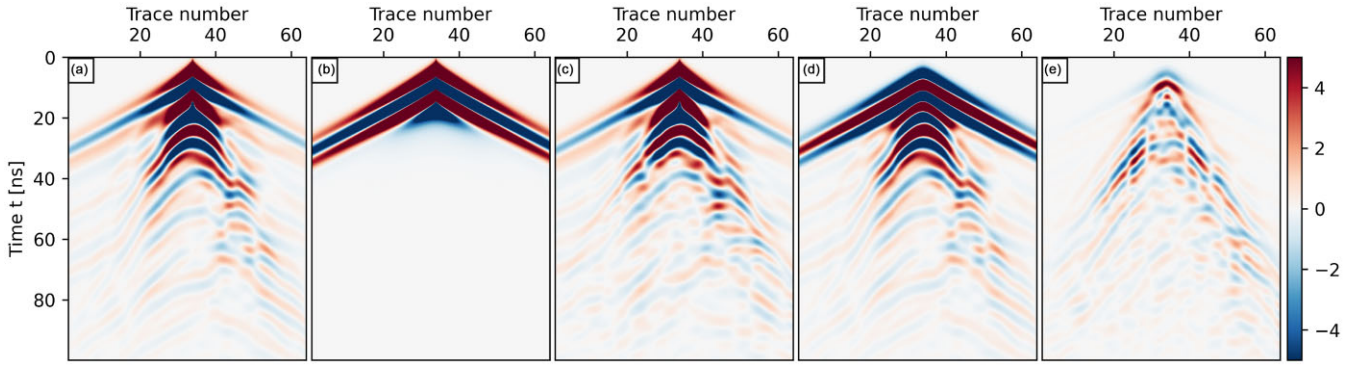


Figure 13. Data comparisons with the source located at $x_s = 6.5$ m using (a) the ground-truth Overthrust models, (b) inaccurate 1D initials in Figs 12(a),(b) and (c) final multiscale FWI models in Figs 12(g) and (h). (d) and (e) denote residuals (a),(b) and (a)–(c), respectively.

3.2.2 Preliminary IFWI

The same network architecture used in the study of the Cross-shaped models, that is a four-layer MLP with 128 neurons in each layer, is adopted in IFWI of the 2-D Overthrust models. Given the inaccurate 1D initial models in Figs 12(a) and (b), IFWI's results of the relative permittivity and conductivity for the Overthrust models using IFWI are plotted in Fig. 14. In comparison with multiscale FWI results in Fig. 12, the inversion results in Fig. 14 demonstrate that IFWI is able to escape from local minima traps and provide detailed variations in relative permittivity and conductivity, as well as precise subsurface structures. In terms of data comparison, Fig. 15 illustrates that, starting from the same initial models, IFWI achieves a better match with the ground-truth shot and produces less data residuals than multiscale FWI, even when compared to multiscale FWI with improved linear initializations as shown in Fig. 11(e). Therefore, it can be said with confidence that the incorporation of deep neural networks with frequency principle significantly reduces the dependence of FWI on the initial model. Furthermore, unlike multiscale FWI that demands the appropriate frequency band selections and complex processing of data for implementations, no manual intervention are required for IFWI.

Despite the success of IFWI in reducing initial model dependence, high-frequency noises are observed in both the IFWI of Cross-shape (Figs 7g and h) and Overthrust models (Fig. 14). As previously discussed, these minor drawbacks stem from the oversensitivity of the current network setting to high-frequency components. In other words, although the network strives to learn a

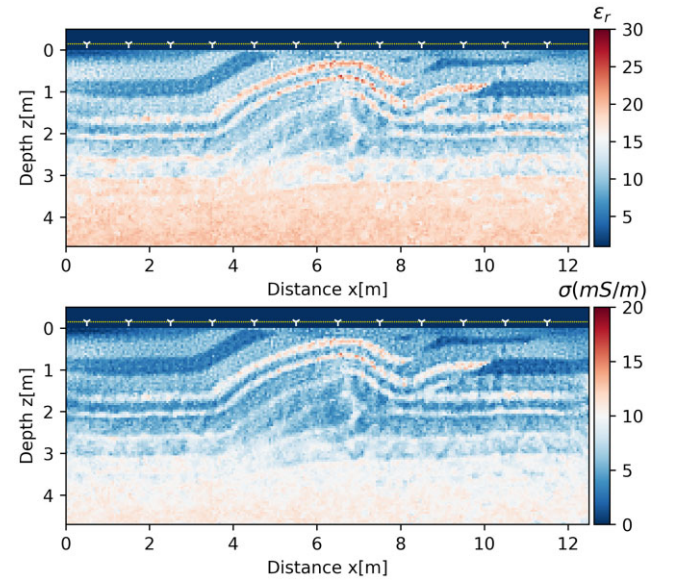


Figure 14. IFWI of the 2D Overthrust models using a four-layer MLP with 128 neurons in each layer and $\omega_0 = 30$, starting from inaccurate 1D initial models shown in Figs 12(a) and (b).

low-frequency function to represent the subsurface relative permittivity and conductivity, the current network settings such as the number of layers, neurons per layer, scaling factor ω_0 and

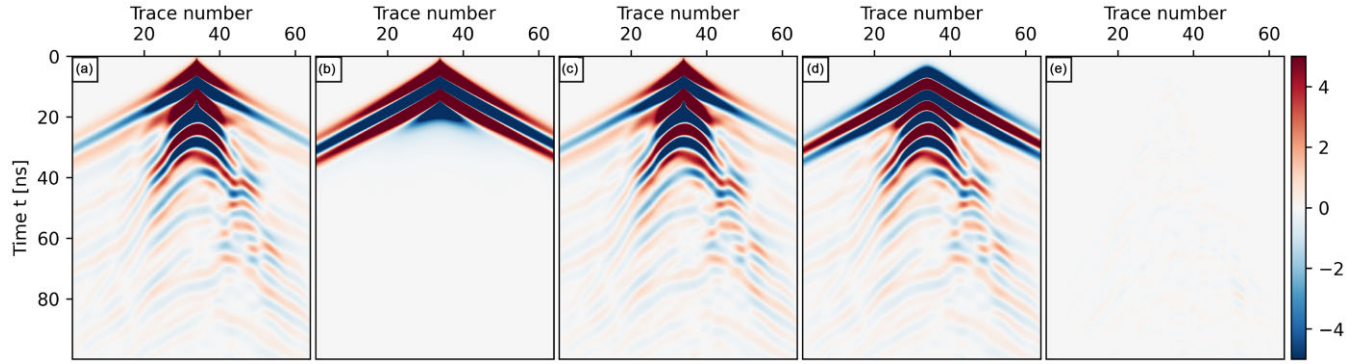


Figure 15. Data comparisons with the source located at $x_s = 6.5$ m using (a) the ground-truth Overthrust models, (b) inaccurate 1D initials in Figs 12(a),(b) and (c) IFWI models in Figs 14(d) and (e) denote residuals (a),(b) and (a)–(c), respectively.

dropout rate p , are too “high-frequency” to accurately represent the subsurface models and adequately suppress these noises. Additionally, in Fig. 14, the relative permittivity and conductivity values inverted by IFWI still exhibit significant differences from their true values. Therefore, understanding how these hyperparameters impact the final results of IFWI is crucial for further improving its accuracy and mitigating these high-frequency noises.

4 DISCUSSIONS

In this section, we delve into IFWI’s behaviour for GPR profiles through numerical comparisons, examining the impact of various hyperparameter settings. While there are four primary hyperparameters in an MLP architecture design utilizing the sinusoidal activation function – including the number of layers, neurons per layer, scaling factor ω_0 and dropout probability rate p – we specifically focus on a fixed four-layer MLP. This choice stems from the fact that a deeper narrow network can often be equivalently represented by a shallower wider network. Additionally, to streamline the hyperparameter analysis and consider the interplay among these parameters, we limit our focus to two specific configurations: a ‘narrow’ MLP with 128 neurons per layer and a relatively ‘wider’ MLP with 256 neurons per layer, referred to as 128×4 and 256×4 MLPs, respectively. The decision to choose only the 128×4 and 256×4 MLPs, rather than wider MLPs with more neurons per layer, is based on their capability to adequately describe high-resolution images with intricate patterns. Consequently, both configurations prove highly effective in representing any 2-D subsurface models, even those with complex and detailed structures. Furthermore, we conduct experiments with different scaling factors ($\omega_0 = [1, 10, 20, 30]$) paired with various dropout rates ($p = [0, 0.1, 0.2, 0.3]$). To provide a more intuitive evaluation of the IFWI results under different hyperparameters, coefficient of determination (R2 Score), structural similarity (SSIM), signal-to-noise ratio (SNR), and mean average percentage error (MAPE) of ground-truth and data simulated using inverted models are selected as metric criteria. However, due to the strong attenuation layer at 3 m depth, these metrics are computed only for the portion of the Overthrust model with depths less than 3 m for comparison.

These four evaluation metrics serve distinct purposes in assessing the performance of the proposed IFWI algorithm. For instance, the R2 score is a statistical measure to assess how well the inverted neural representation model (i.e. $\mathcal{N}_\Theta$) fits the ground-truth subsurface parameters $\mathbf{m}$. Specifically, it indicates the proportion of the

variance in target variables (i.e. the ground-truth subsurface parameters $\mathbf{m}$) that is explained by the predicted variables (i.e. the inverted subsurface parameters $\hat{\mathbf{m}} = \mathcal{N}_\Theta(\mathbf{x})$). The mathematical expression of R2 score used in this paper is written as

$$R2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \quad (7)$$

where, $SS_{\text{res}} = \sum \|\tilde{\chi} - \chi\|^2$ and $SS_{\text{tot}} = \sum \|\chi - \bar{\chi}\|^2$. $\tilde{\chi}$ represents one of the inverted subsurface parameters in $\mathbf{m}$ (specifically, either ϵ_r or σ in mS m^{-1}), with χ indicating its corresponding ground-truth. $\bar{\chi}$ denotes the average value of χ . An R2 of 1 means perfect prediction, while an R2 of 0 means the model does not explain any of the variance.

On the other hand, the SSIM is a perceptual metric used to measure the structural similarity between the inverted subsurface model and its corresponding ground-truth. The SSIM of images x and y can be calculated as

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (8)$$

where, μ_x, μ_y are the mean values, σ_x^2, σ_y^2 are variances, σ_{xy} is covariance, and C_1, C_2 are small constants to stabilize the division. In this study, we adopt the default settings for these parameters, with $C_1 = 0.01$, $C_2 = 0.03$. As indicated, the SSIM values range from -1 to 1 , with a value of 1 indicating perfect similarity.

In addition to R2 and SSIM, the SNR is adopted to assess the high-frequency noises introduced in the IFWI results. The mathematical formulation of SNR used in this paper is written as

$$SNR = 20 \times \log \frac{\sum \tilde{\chi}^2}{\sum \|\tilde{\chi} - \chi\|^2} \quad (9)$$

where the definitions of $\tilde{\chi}$ and χ in equations (9) and (10) are same as those provided in equation (7).

Furthermore, the MAPE is selected to evaluate the overall data accuracy of the inverted neural representation model by expressing the error as a percentage of the ground-truth values. The mathematical formulation of MAPE is written as

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\mathbf{d}^i - \mathbf{d}_{\text{obs}}^i}{\mathbf{d}_{\text{obs}}^i} \right| \times 100\% \quad (10)$$

where $\mathbf{d}_{\text{obs}}$ denotes the ground-truth observations, $\mathbf{d}^i$ represents the simulated data using the inverted models, and n denotes number of elements in $\mathbf{d}$ for mean-value computation.

Fig. 16 shows IFWI results obtained using a 128×4 MLP with a zero dropout rate ($p = 0$) and various scaling factors

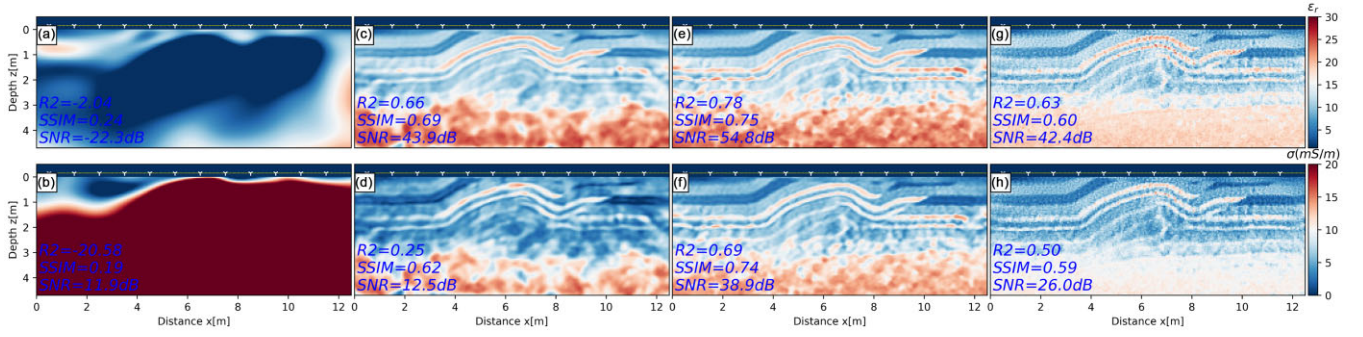


Figure 16. IFWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row using a 128×4 MLP with a zero dropout rate ($p = 0$) and various scaling factors. (a) and (b): $\omega_0 = 1$; (c) and (d): $\omega_0 = 10$; (e) and (f): $\omega_0 = 20$; (g) and (h): $\omega_0 = 30$ (same with Fig. 14).

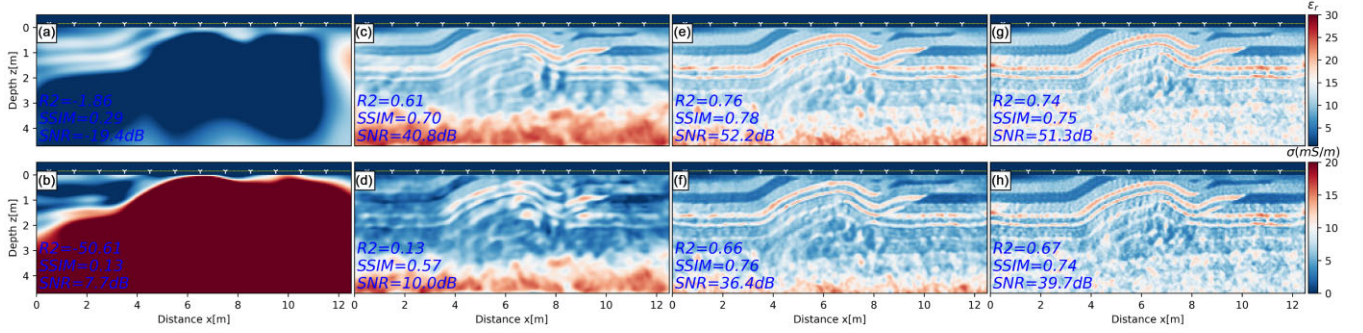


Figure 17. IFWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row using a 256×4 MLP with a zero dropout rate ($p = 0$) and various scaling factors. (a) and (b): $\omega_0 = 1$; (c) and (d): $\omega_0 = 10$; (e) and (f): $\omega_0 = 20$; (g) and (h): $\omega_0 = 30$.

($\omega_0 = [1, 10, 20, 30]$) from left to right. The results in Figs 16(a) and (b) indicate that IFWI using a 128×4 MLP with a very small scaling factor ($\omega_0 = 1$) only captures the global trend of the relative permittivity and conductivity, without representing any detailed structures. This implies that a very small scaling factor may compromise the MLP's capability to depict high-frequency details, rendering it inadequate for complex subsurface media characterization. Figs 16(c)–(h) illustrate the increasing capacity of the MLP to delineate detailed structures as ω_0 increases, consequently enhancing the subsurface reconstruction in IFWI. However, it is crucial to acknowledge that excessive sensitivity to high-frequency components in the MLP, especially with larger ω_0 , may lead to the introduction of high-frequency noise in IFWI results, as evidenced in Figs 16(g) and (h).

Following the analysis presented above, similar IFWI experiments were conducted using a 256×4 MLP with various scaling factors, and the results are depicted in Fig. 17. The first column of Fig. 17 suggests that while increasing the number of neurons per layer with a very small scaling factor may marginally improve the MLP's capacity to represent complex structures compared to Figs 16(a) and (b), IFWI still fails to provide useful information regarding subsurface anomalies in terms of relative permittivity and conductivity. Regarding the relatively wider MLP (256×4), we also observe an enhanced ability for detailed structure representation as the scaling factor ω_0 increases. However, when compared to a narrow MLP 128×4 with the same scaling factor and a zero dropout rate, a wider MLP may slightly decrease IFWI's accuracy in the inversion of deep layers.

Fig. 18 presents a comparison of metrics for IFWI using both 128×4 and 256×4 MLPs, with a zero dropout rate and varying scaling factors. The results indicate that increasing the scaling

factor appropriately can enhance the performance of IFWI and reduce data misfits. However, excessively large scaling factors may decrease the SNR of IFWI results by introducing high-frequency noises. Considering all above metrics, when the dropout rate is zero, IFWI with a scaling factor of 20 demonstrates the best performance for both 128×4 and 256×4 MLPs. These findings are consistent with our previous intuitive analysis presented in Figs 16 and 17.

We also conducted IFWI experiments using both 128×4 and 256×4 MLPs, maintaining a fixed scaling factor of $\omega_0 = 30$, while varying dropout rates to assess their impact on IFWI. The results presented in Figs 19 and 20 reveal that with a large scaling factor, introducing dropout neurons leads to slight improvements in the values of parameters in the shallow layers. However, high-frequency noises are only marginally mitigated due to the oversensitivity of the sine function when coupled with a large scaling factor. Additionally, the metrics comparison in Fig. 21 demonstrates that augmenting the dropout rate does not yield enhancements in IFWI outcomes when employing a large scaling factor. Furthermore, we observe a near-linear increase in data misfits with increasing dropout rates, particularly noticeable in IFWI utilizing the narrow MLP. Conversely, the data misfit of IFWI employing the wider MLP exhibits a slower rate of growth.

Next, the scaling factor is fixed at its optimal value of $\omega_0 = 20$ from IFWI experiments with a zero dropout rate, based on the analysis of Fig. 18. IFWI experiments are then conducted using this fixed scaling factor ($\omega_0 = 20$) with varying dropout rates to further investigate the potential for improving their performance. Compared to IFWI experiments with a scaling factor of 30 (Figs 19 and 20), incorporating dropout regularization for IFWI with a scaling factor of 20 leads to improved mitigation of high-frequency noises and

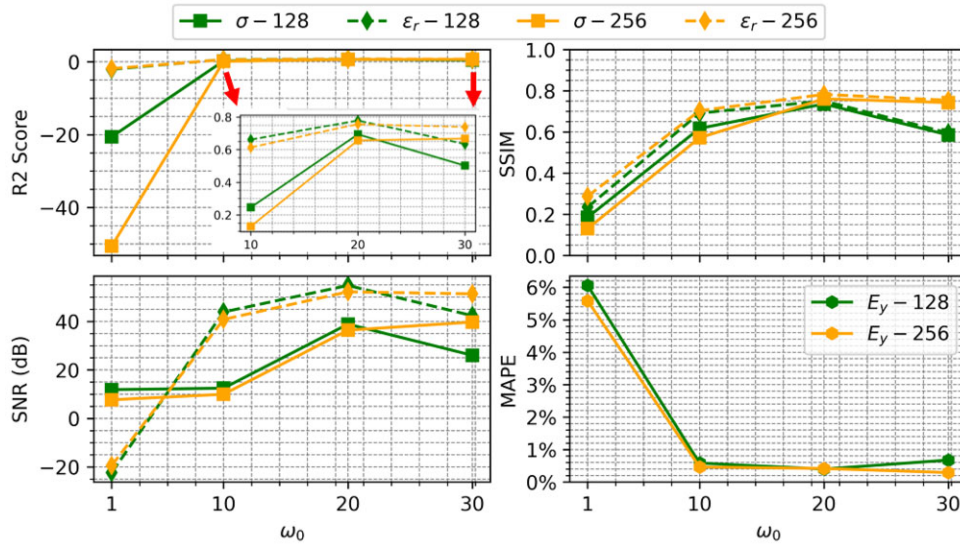


Figure 18. Metrics comparison of IFWI using MLPs with a zero dropout rate and various scaling factors.

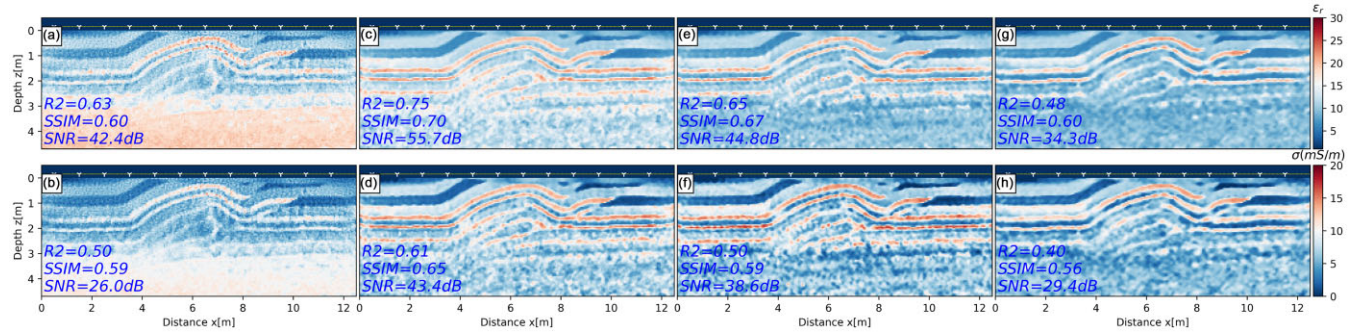


Figure 19. IFWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row using a 128×4 MLP with a fixed scaling factor $\omega_0 = 30$ and various dropout rates. (a) and (b): $p = 0$; (c) and (d): $p = 0.1$; (e) and (f): $p = 0.2$; (g) and (h): $p = 0.3$.

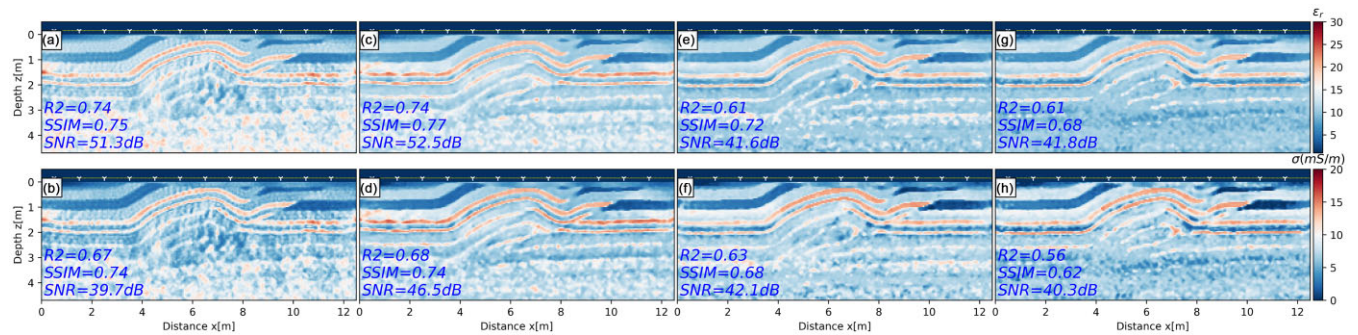


Figure 20. IFWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row using a 256×4 MLP with a fixed scaling factor $\omega_0 = 30$ and various dropout rates. (a) and (b): $p = 0$; (c) and (d): $p = 0.1$; (e) and (f): $p = 0.2$; (g) and (h): $p = 0.3$.

enhanced accuracy of shallow layers, as shown in Figs 22 and 23. Comparing IFWI results with the same dropout rates in Figs 22 and 23 indicates that increasing the width of the MLPs can enhance IFWI's reconstruction in depth to a certain extent, as highlighted by the red-dashed boxes. Similarly, contrasting the performance across different dropout rates in these figures suggests that appropriately augmenting the dropout rate can also enhance IFWI's performance in deeper layers, particularly for the IFWI with the wider MLP, as illustrated by the red arrows in Fig. 23. The metrics comparison for

both narrow and wider MLPs with a fixed scaling factor of 20 and varying dropout rates is depicted in Fig. 24. It is observed that with increasing dropout rates, IFWI with the wider MLP exhibits a slight advantage across all four metric criteria.

In summary, the hyperparameter analysis outlined above highlights the significant influence of the scaling factor on the MLP's representation ability. It is observed that an MLP coupled with a very small scaling factor ($\omega_0 = 1$) may inadequately capture detailed features. Therefore, we recommend a relatively larger scaling

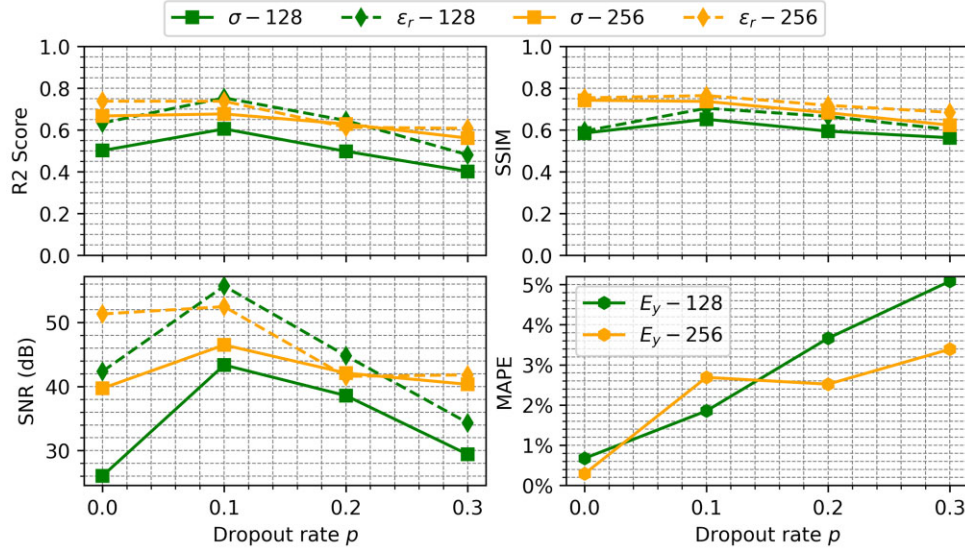


Figure 21. Metrics comparison of IFWI using MLPs with a fixed scaling factor $\omega_0 = 30$ and various dropout rates.

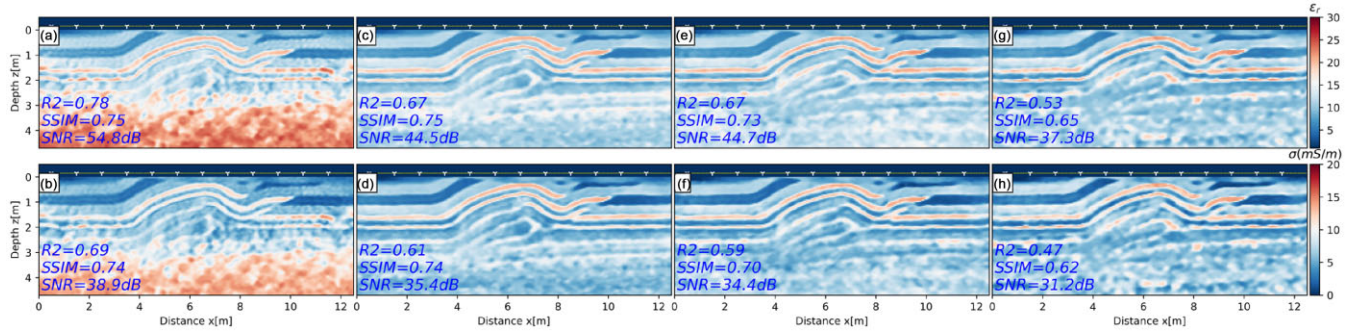


Figure 22. IFWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row using a 128×4 MLP with a fixed scaling factor $\omega_0 = 20$ and various dropout rates. (a) and (b): $p = 0$; (c) and (d): $p = 0.1$; (e) and (f): $p = 0.2$; (g) and (h): $p = 0.3$.

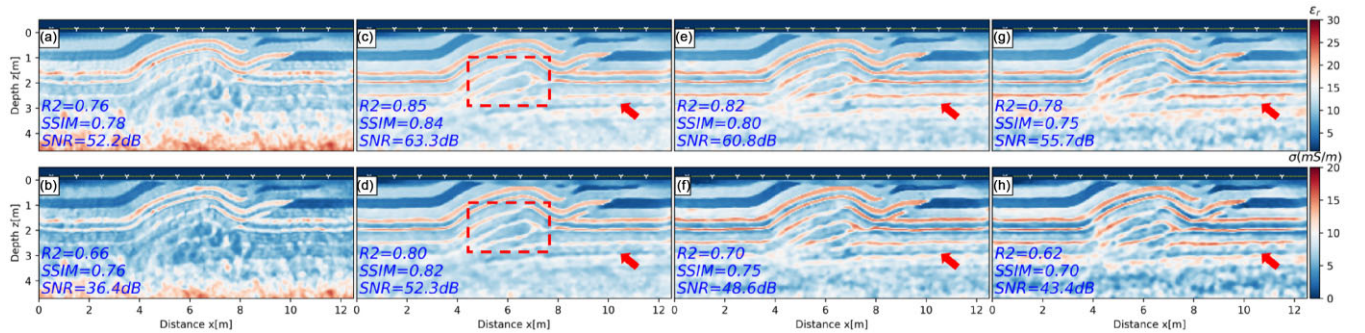


Figure 23. IFWI of the Overthrust model with relative permittivity in the top row and conductivity in the bottom row using a 256×4 MLP with a fixed scaling factor $\omega_0 = 20$ and various dropout rates. (a) and (b): $p = 0$; (c) and (d): $p = 0.1$; (e) and (f): $p = 0.2$; (g) and (h): $p = 0.3$.

factor, such as $\omega_0 = 30$ or 20 , as the initial hyperparameter for IFWI. Comparing narrow and wider MLPs reveals similar performance in shallow-to-middle layers, with a slight advantage for the wider MLP in depicting middle-depth layer features. In such scenarios, focusing solely on minimizing data misfit may not yield optimal inversion outcomes. For instance, in Figs 21 and 24, both IFWIs with a zero dropout rate $p = 0$ achieve the minimal data misfit; however, their corresponding inversion results (Figs 22a,b

and 23a,b) for shallow-to-middle layers are not optimal due to the presence of the high-frequency noise. Both gradually reducing the scaling factor and introducing dropout neurons can effectively mitigate high-frequency noise, however, the choice between them depends on the depth of the interest zone. For the interest zone of shallow-to-middle layers, a wider network with dropout neurons is recommended as its slight advantages in depicting middle-depth layers. Consequently, for the interest zone of deep

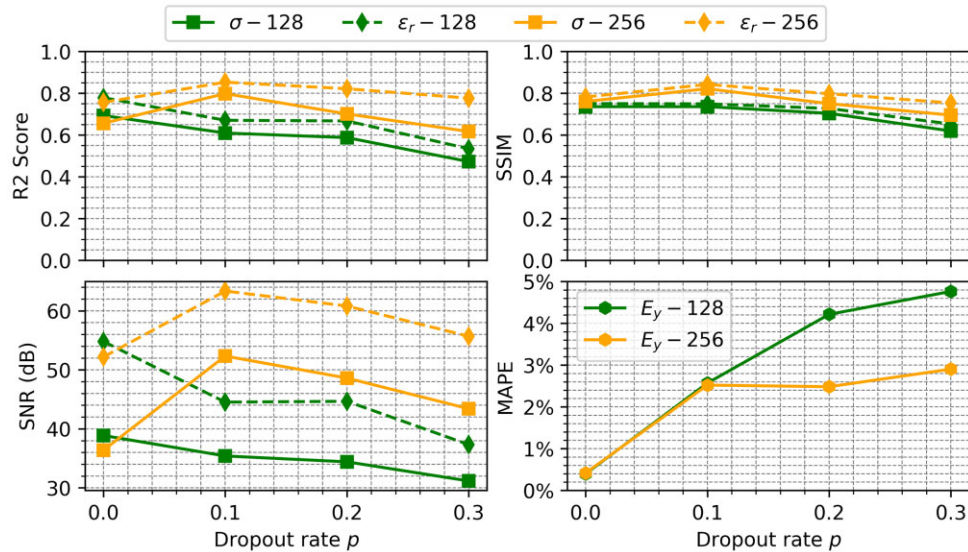


Figure 24. Metrics comparison of IFWI using MLPs with a fixed scaling factor $\omega_0 = 20$ and various dropout rates.

layers, it is advisable to gradually decrease the scaling factor in a narrow MLP until IFWI delivers the desired results, as introducing dropout neurons or employing a wider MLP may compromise IFWI's accuracy in deep layers. It is noted that the crosstalk effects in multiparameter FWI are not discussed in this paper. However, the proposed IFWI framework may offer a potential solution to mitigate these effects, particularly by incorporating the encoder-decoder neural representation discussed in Dhara and Sen (2023).

5 CONCLUSIONS

A key approach to characterizing the physical properties of the subsurface is GPR with appropriate data processing, where FWI is one of the leading techniques to fully exploit the acquired GPR profile and reconstruct high-resolution subsurface image. However, due to the limited bandwidth and illumination of local GPR measurements, FWI of GPR data is an ill-posed optimization task with a high degree of non-linearity. That is, FWI suffers greatly from local minima problems and its performance is highly dependent on the quality of the initial model. In this paper, we propose the implicit full waveform inversion for GPR, in which the subsurface parameters are delineated as an implicit and continuous function using deep neural network representation, to reduce the dependence of FWI on the initial model. By integrating deep neural networks into FWI, IFWI can be viewed as training a physics-constrained neural network, which may adhere to frequency principle of prioritizing the learning of low-frequency information in deep learning optimization. In essence, IFWI prioritizes acquiring a low-frequency representation of subsurface parameters before reconstructing high-frequency fine-grained structures. This characteristic renders it an automated multiscale inversion strategy and ensures its potential for reduced reliance on initial models, while eliminating the need for manual frequency band manipulation.

Two numerical experiments are used to examine the effectiveness and performance of IFWI. The results show that IFWI is able to finely reconstruct subsurface images and anomalies in the presence of suboptimal initial models, without manual manipulation of the usage of frequency components. In contrast, both FWI and multiscale FWI with selected frequency bands fail to provide useful

information about the subsurface due to being trapped in local minima. It can therefore be concluded that the IFWI is less dependent on the quality of initial model. In addition, comprehensive analysis of hyperparameters, including the number of neurons per layer, the scaling factor ω_0 , and the dropout rate, is conducted to investigate the influence of network architecture on IFWI's performance. These comparisons suggest that a relatively larger scaling factor is recommended in initial attempts of IFWI as it ensures the network's ability to represent fine-grained structures. Both reducing the scaling factor and introducing dropout neurons are effective strategies for mitigating high-frequency noise in inversion results. However, the choice between a wider network with dropout neurons and a narrower network with a reduced scaling factor depends on the depth of the target zone. For areas not in deep layers, a wider network with dropout neurons is recommended. Conversely, if the target zone is in deep layers, it is advisable to opt for a narrower network with a reduced scaling factor as introducing dropout neurons or employing a wider network in deep layers may compromise IFWI's accuracy.

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DATA AVAILABILITY

No new data were generated or analysed in support of this research.

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